

HUM: Hear, Understand, Match for Unseen Audio-Text Alignment

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Abstract—Automatic metrics for text-to-audio generation still agree poorly with human judgments of whether a clip actually matches its prompt. The ICASSP 2027 XACLE Grand Challenge asks for a model that predicts the mean human alignment score for an (audio, prompt) pair, and its 2027 edition tests on audio from generation systems never seen in training. We introduce HUM (Hear, Understand, Match), a trainable network that hears a clip through frozen audio and CLAP encoders, understands it with a QLoRA-tuned Qwen2.5-Omni audio-language model prompted as a yes/no verifier, and matches both signals through a gating network and an ordinal score head. We target the unseen-generator shift directly with a within-prompt pairwise ranking loss that compares clips generated for the same prompt, which cancels prompt difficulty and rewards orderings that do not depend on the generator. We evaluate with a leave-one-generator-out protocol that reports the seen-to-unseen correlation gap for every baseline and for HUM, with bootstrap confidence intervals. Through this approach, we aim to obtain a statistically significant reduction of this gap relative to a strong CLAP-regression baseline.

Index Terms—Text-to-audio evaluation, audio-language models, learning to rank, domain generalization, parameter-efficient fine-tuning.

I. INTRODUCTION

A. Motivation

Text-to-audio (TTA) models such as AudioLDM and Tango can turn a sentence into sound, but checking whether the sound actually matches the sentence still requires human listening tests, which are slow, costly, and hard to reproduce. The standard automatic proxy, CLAPScore, correlates weakly with human relevance ratings: on the RELATE dataset, LAION-CLAP reaches a Spearman rank correlation (SRCC) of only 0.280 [1], [2], and contrastive audio-text models struggle with event order and attribute binding [3]. A learned evaluator that agrees with listeners would make TTA benchmarking cheaper and could serve as a reward signal for better generators.

B. Problem Statement

Given generated audio a and its prompt t , we predict a score $\hat{y}$ whose ranking over a test set matches the mean human alignment rating $y \in [0, 10]$. XACLE ranks systems by SRCC, with linear correlation (LCC), Kendall’s tau (KTAU), and mean-squared error (MSE) as tie-breaks [4]. The 2027 edition adds an *unseen-generation-system* setting: test audio

comes from generators that are absent from training [5]. A model can therefore look strong on validation by learning generator-specific artifacts and still fail on the test set, which makes the task a domain-generalization problem rather than only a regression problem.

C. Significance

Every 2026 system optimized correlation under seen generators. The official baseline (BYOL-A + RoBERTa + LSTM) reached 0.3345 test SRCC [4], while the strongest systems reached 0.61–0.64 [6]–[8]. Since new TTA models appear every few months, a practical evaluator must generalize to generators it has never heard. At the same time, a zero-shot audio-language model judge already reaches 0.556 SRCC on RELATE [9], so strong in-domain correlation is an engineering target; generalization to unseen generators is the open research question.

D. Goals and Contributions

We separate engineering targets from the research bet:

- G1.** Reproduce the official baseline (B1) and our CLAP-regression fallback (B4) by Checkpoint I (Oct. 20).
- G2.** *Engineering*: reach in-domain validation SRCC at least equal to zero-shot AQAScore [9].
- G3.** *Research bet*: reduce the leave-one-generator-out gap Δ (Eq. (7)) relative to B4, with a 95% bootstrap interval that excludes zero across at least three folds.
- G4.** Submit a rule-compliant system to XACLE 2027 by Dec. 3
- G5.** Release a reproducible repository, a Gradio demo, and a demonstration video.

Our contributions are (i) a within-prompt ranking objective for generator-invariant alignment scoring, (ii) an unseen-generator benchmark protocol that reports Δ for re-implemented baselines, and (iii) a single-network judge that combines language-model verification with embedding similarity.

II. RELATED WORK

A. Embedding-Based and Learned Metrics

CLAPScore [10] measures how well audio and text match by comparing their CLAP embeddings. Human-CLAP improves CLAP by fine-tuning it on human ratings [1], while

M2D-CLAP learns stronger general-purpose audio-language features [11]. RELATE released human relevance scores for TTA audio [2], and XACLE 2026 built its benchmark on this lineage [4].

The top XACLE 2026 systems extended these ideas. MoEScore gated several CLAP-based experts together with a BEATs [12]–RoBERTa cross-attention expert, reaching an SRCC of 0.6402 [6]. The third-place TMU system fine-tuned a compact audio-language model using CLAP pseudo-labels and a listwise ListNet loss [13], reaching 0.632 [7]. This ListNet loss ranks clips across each training batch without regard to which prompt each clip came from; HUM additionally applies a ranking loss restricted to clips that share the same prompt, so the model learns which clip best matches a specific prompt rather than which prompts are generally easier to score. SPO-CLAPScore standardized each listener’s ratings, reaching 0.6142 [8].

HUM borrows gating, ranking, and listener standardization from these systems. Unlike them, HUM also targets generalization to audio-generation systems not seen during training, which is a problem none of these systems needed to solve, since XACLE 2026 tested only on seen generators.

B. Audio-Language Models as Judges

AQAScore asks Qwen2.5-Omni [14] whether a clip contains the events described in its prompt, using the probability of a “Yes” answer as the score; zero-shot, it reaches SRCC 0.556 on RELATE, above similarity-based metrics [9].

HUM fine-tunes this verifier with QLoRA [15], [16], rather than using it zero-shot, and fuses its output with an embedding-based signal instead of relying on the language model alone.

C. Generalization to Unseen Systems

Evaluating generative models faces challenges when metrics overfit to generator-specific artifacts rather than true semantic alignment [17]. In subjective speech and audio evaluation, challenges such as AudioMOS 2025 have shown performance degradation when models evaluate synthetic samples generated by unseen synthesis engines or vocoders [18]. A common fail point in cross-generator evaluation is shortcut learning, where evaluators learn system-specific acoustic cues instead of semantic alignment [17].

To counter this bias, domain generalization methods encourage the network to learn features that remain invariant across different audio generators. Specifically, domain-adversarial training with a Gradient Reversal Layer (GRL) [19] trains the model to score alignment without recognizing the source generator. Beyond training-time invariance, test-time adaptation methods like Tent [20] can adjust predictions directly on unlabeled test batches. HUM applies these concepts to construct a generator-invariant latent space $\mathbf{z}$, reducing the drop in rank correlation (Δ) when evaluating unseen synthesis models. Its primary invariance signal, however, is the within-prompt ranking loss, which needs no generator labels; the GRL branch and test-time calibration are add-ons that are kept only if they reduce Δ .

III. PROPOSED APPROACH

A. Overview

HUM is one network with three stages (Fig. 1). **Hear:** the clip passes through the frozen Qwen2.5-Omni audio encoder and frozen M2D-CLAP audio and text encoders, whose outputs are computed once and cached. **Understand:** the Qwen2.5-Omni Thinker, loaded with a 4-bit base and trainable LoRA adapters (the speech-output Talker is removed), receives the audio tokens and the query “Does this audio contain the sound events described by ‘ t ’?” It returns p_{yes} , the normalized probability of “Yes” versus “No,” and $\mathbf{h}_s$, the hidden state at a dedicated score token. **Match:** a gating MLP fuses $\mathbf{h}_s$, p_{yes} , and the CLAP cosine similarity s_{clap} into a shared representation $\mathbf{z}$, and an ordinal head predicts probabilities p_k over $K = 21$ bins $c_k \in \{0, 0.5, \dots, 10\}$. The score is the expected bin value:

$$\hat{y} = \sum_{k=1}^K p_k c_k. \quad (1)$$

The 2026 rules prohibited ensembling results from multiple models but allowed internal ensembling within a single model [4]. HUM is one network, so it stays compliant if the 2027 rules are the same or stricter.

B. Within-Prompt Ranking

XACLE 2026 provided three clips per prompt [8], which may come from different generators; we will confirm this on the 2027 release. For clips i and j of the same prompt with human scores $y_i > y_j$, we add a margin ranking loss:

$$\mathcal{L}_{\text{pair}} = \frac{1}{|\mathcal{P}|} \sum_{(i,j) \in \mathcal{P}} \max(0, m - (\hat{y}_i - \hat{y}_j)), \quad (2)$$

where $\mathcal{P}$ is the set of within-prompt pairs whose score difference exceeds a noise threshold and m is a margin. Because both clips share the prompt, prompt difficulty cancels, and the model must rank clips by how well each one realizes the same text. We hypothesize that this reduces reliance on generator artifacts without requiring generator labels. Mini-batches are built so that all clips of a prompt appear together.

C. Generator Invariance

Learned evaluators often overfit to artifacts from specific audio generators, leading to poor generalization on unseen systems [17]. To prevent this, HUM adds an adversarial branch to the shared multimodal representation $\mathbf{z}$ to filter out generator-specific information.

During training, $\mathbf{z}$ passes through a Gradient Reversal Layer (GRL) [19] into a classifier that predicts the source generator:

$$\mathcal{R}(\mathbf{z}) = \mathbf{z}, \quad \frac{\partial \mathcal{R}(\mathbf{z})}{\partial \mathbf{z}} = -\mathbf{I}. \quad (3)$$

The classifier is a two-layer MLP that outputs probability distributions over known generator classes. When generator IDs are available, the branch trains using standard cross-entropy loss $\mathcal{L}_{\text{adv}}$. If generator labels are missing, we compute acoustic-residual embeddings by removing prompt-aligned

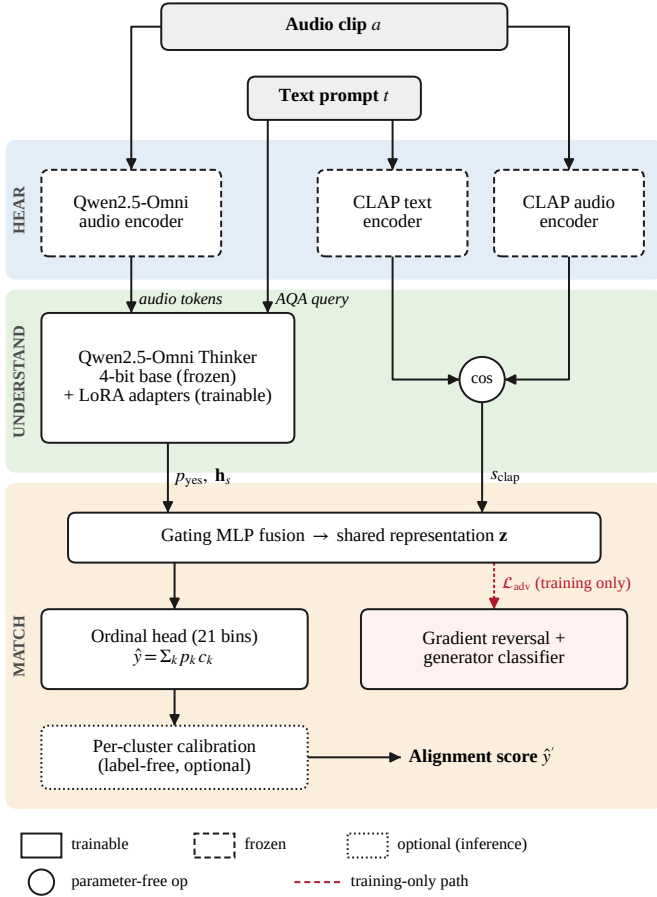


Fig. 1. HUM architecture. *Hear*: frozen audio and CLAP encoders. *Understand*: LoRA-tuned Qwen2.5-Omni verifier. *Match*: gated fusion, ordinal score head, and a training-only gradient-reversal generator classifier. Solid borders denote trainable modules; dashed borders denote frozen modules.

CLAP components and cluster them using spherical K -means to generate pseudo-domain labels.

To prevent adversarial gradients from destabilizing early training, the loss weight $\gamma(\tau)$ increases smoothly from 0 to γ_{max} according to the sigmoid ramp schedule in (5).

D. Training Objective

The full objective is

$$\mathcal{L} = \mathcal{L}_{\text{rank}} + \alpha \mathcal{L}_{\text{ord}} + \beta \mathcal{L}_{\text{mse}} + \delta \mathcal{L}_{\text{pair}} + \gamma(\tau) \mathcal{L}_{\text{adv}} \quad (4)$$

where $\mathcal{L}_{\text{rank}}$ is ListNet over the mini-batch [13], which directly rewards correct orderings and matches the rank-based SRCC; $\mathcal{L}_{\text{ord}}$ is the KL divergence from a Gaussian-softened target over the 21 bins ($\sigma = 0.5$), a soft-bin strategy used by top AudioMOS 2025 systems [18]; and $\mathcal{L}_{\text{mse}}$ on $\hat{y}$ stabilizes the score scale for the LCC and MSE tie-breaks. The weights α, β, δ , and m are tuned on validation data, and the adversarial weight follows

$$\gamma(\tau) = \gamma_{\text{max}} \left(\frac{2}{1 + e^{-10\tau}} - 1 \right), \quad (5)$$

where $\tau \in [0, 1]$ is training progress. The default system sets $\gamma_{\text{max}} = 0$; with only a handful of generators the adversarial

signal is weak, and clusters may capture content or loudness rather than generator style, so the GRL branch is kept only if it passes the acceptance rule in Section III-G.

E. Training Stages

S0 caches the frozen audio-encoder and CLAP outputs. **S2**, the default system, fine-tunes on XACLE with $\mathcal{L}_{\text{rank}}$, $\mathcal{L}_{\text{ord}}$, $\mathcal{L}_{\text{mse}}$, and $\mathcal{L}_{\text{pair}}$. Three add-ons run only if they pass the acceptance rule: **S1**, a warm-up on declared external caption-audio pairs labeled by a CLAP teacher, which was the main performance driver for TMU [7]; **S3**, which enables the GRL branch; and per-listener score standardization [8], if listener IDs are released.

F. Optional Test-Time Calibration

Evaluating audio from unseen generators often introduces systematic score offsets due to differing baseline acoustic characteristics. To correct for these shifts without requiring ground-truth ratings at test time, we use a label-free calibration step in the spirit of test-time adaptation [20].

Given an unlabelled test split, we cluster the audio representations into C acoustic groups using K -means. For each cluster c , we compute its mean predicted score μ_c alongside the overall test set mean μ . We then adjust the raw ordinal score $\hat{y}$ using:

$$\hat{y}' = \hat{y} - \lambda (\mu_c - \mu), \quad \lambda \in [0, 1]. \quad (6)$$

The hyperparameter λ controls correction strength and is tuned on validation Leave-One-Generator-Out (LOGO) splits to maximize rank correlation (SRCC). Setting $\lambda = 0$ disables calibration, preserving raw network outputs if cluster separations are weak. Because a single monotonic map applied to all predictions cannot change SRCC, the correction must act per cluster; full equalization could also erase genuine quality differences between generators, so λ defaults to 0. The step keeps model weights frozen and is applied only if the 2027 rules allow transductive use of the evaluation audio.

G. Baselines, Metrics, and Evaluation Protocol

Baselines: B1 is the official 2027 baseline; B2 is zero-shot CLAPScore; B3 is zero-shot AQAScore [9]; and B4 is an MLP on frozen M2D-CLAP features trained with the same losses, which is our guaranteed submission. **Metrics:** we report SRCC (primary), LCC, KTAU, MSE, and per-generator SRCC, with 95% bootstrap intervals from 1,000 resamples, because 2026 placements were separated by as little as 0.0001 SRCC [8]. **Protocol:** all splits are grouped by prompt, so clips of one prompt never straddle training and validation. Leave-one-generator-out (LOGO) cross-validation trains on all but one generator (or pseudo-domain) and tests on the held-out one. The headline number is

$$\Delta = \text{SRCC}_{\text{seen}} - \text{SRCC}_{\text{unseen}}. \quad (7)$$

We report Δ for B2–B4 and HUM, which forms our benchmark contribution. **Acceptance rule:** a component is kept only if it lowers Δ without dropping in-domain SRCC below B4.

Ablations: each loss term removed; with and without S1, GRL, calibration, and listener standardization; zero-shot versus fine-tuned verifier; and 3B versus 7B.

IV. SYSTEM DESIGN

A. Architecture and Workflow

The pipeline has seven steps. First, we ingest XACLE and any declared external data, check that counts and score ranges are correct, and remove RELATE clips that duplicate XACLE audio or prompts to prevent data leakage. Second, we cache frozen features and build prompt-grouped LOGO (leave-one-generator-out) folds. Third, we train the baseline model B4, followed by HUM. Fourth, we evaluate every run using one shared metrics script. Fifth, we select the final configuration using our acceptance rule. Sixth, we run frozen inference on the evaluation release. Seventh, we export predictions in the official submission format. Every result can be reproduced from a config file and a fixed random seed.

B. Data Sets

Our primary data is the XACLE 2027 training and validation set, scheduled for release on September 24, 2026. For reference, XACLE 2026 included 7,500 training pairs, 3,000 validation pairs, and 3,000 test pairs, each rated on a 0–10 scale [4]; we assume a similar scale for 2027. RELATE [2] serves as our declared external dataset, and AudioCaps and Clotho provide caption–audio pairs for the optional S1 stage. All external data and pretrained models we use will be declared to the challenge organizers.

C. Languages, Tools, and Compute

We build HUM using Python, PyTorch, Hugging Face Transformers and PEFT [15], bitsandbytes [16], FlashAttention-2, torchaudio, scikit-learn, SciPy, Weights & Biases, Gradio, and GitHub.

Compute is split across two platforms. Google Colab (A100, 40/80 GB) runs our 7B QLoRA experiments [16] in bf16 with gradient checkpointing, a batch size of 2–4 with gradient accumulation, and checkpoints saved to Google Drive every 200 steps so training can resume after Colab session limits. Since the audio encoder stays frozen, we cache its outputs once, so each training epoch only needs to run the language model. VT GLogin handles CLAP feature caching, B4 training, LOGO sweeps, and ablation studies. To keep this manageable, LOGO sweeps and ablations run on B4 or the 3B model, and the 7B model is reserved for a few decisive runs; in week one we also measure GLogin queue times, and if they are too long, the sweeps move to Colab.

We estimate 1–2 A100-hours per 7B training epoch and budget roughly 60 A100-hours in total, with actual throughput measured in week one. If no 7B run finishes by October 19, we fall back to the 3B model or to B4 alone. Qwen2.5-Omni-7B [14] uses an Apache-2.0 license; the 3B fallback uses the non-commercial Qwen Research License, so we will confirm it is approved for the 2027 challenge before relying on it.

D. Demo and Usability

We will provide an interactive web demo built with Gradio to evaluate user-provided audio and text inputs. The application accepts an audio file (WAV or MP3) and a text prompt, runs inference through the HUM pipeline, and outputs:

- 1) The final alignment score $\hat{y}$ on a continuous scale.
- 2) The predicted probability distribution across the 21 ordinal bins to indicate scoring confidence.
- 3) The gating fusion weights, showing the relative contribution of the Qwen2.5-Omni verifier versus direct CLAP similarity.
- 4) An interactive toggle comparing raw scores $\hat{y}$ with cluster-calibrated scores $\hat{y}'$.

Using 4-bit base-model quantization, we will report measured GPU memory and per-sample latency for the demo configuration.

E. Risks and Fallbacks

If 7B training proves too slow, we fall back to the 3B model or submit B4 alone. If the pairwise loss does not reduce the generalization gap (Δ), we report this as a negative result with confidence intervals, which still supports the benchmark contribution. If generator IDs are not released, we use pseudo-domain labels or drop GRL [19], since the pairwise loss needs no such labels. If the final 2027 rules differ from these assumptions, we will re-scope the project within 48 hours of the release.

ACKNOWLEDGMENT

AI tools were used to help draft and edit this proposal and will assist with coding; all technical choices, claims, and citations were reviewed by the team.

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APPENDIX A TASK ASSIGNMENT FOR EACH MEMBER

TABLE I
TASK ASSIGNMENT FOR EACH MEMBER

Member	Primary Responsibilities
[Leader] Afeef Ali	Leading overall coordination and setting up the Qwen2.5-Omni Thinker with QLoRA. Responsible for designing the AQA prompt and score token, building the ordinal head, implementing the Eq. (4) loss function, and running Stage S2 training.
Anthony Merino	Handling M2D-CLAP feature caching and building the CLAP-regression fallback (B4). Also developing the gating fusion MLP, an optional S1 pseudo-label warm-up, and listener standardization.
Erdem Yilmaz	Defining generator domains (via IDs or prompt-residual K-means), building the GRL head with its γ schedule for Stage S3, and handling per-cluster calibration and λ tuning.
Kean Jaldin Guzman	Managing data ingestion, validation, and RELATE de-duplication. Setting up prompt-grouped LOGO folds, the metrics and bootstrap script, Slurm jobs, and the ablation runner, plus formatting the final submission.
All	Weekly syncs, code review on every pull request, a rule check after the Sept. 24 release, and presentations.

APPENDIX B SCHEDULE

TABLE II
PROJECT SCHEDULE (FALL 2026)

Week (2026)	Tasks	Milestone / Deliverable
Sep 15–22	Form team, elect leader; survey XACLE 2026 papers; draft proposal; set up repo and web page	Sep 22: Proposal due
Sep 23–29	Sept. 24 data + baseline release: download; verify rules, metric, generator/listener IDs; data validation; re-scope within 48 h	Rules/data checklist
Sep 30–Oct 6	Run official baseline (B1), zero-shot CLAPScore (B2) and AQAScore (B3); cache CLAP features; build LOGO folds	Baseline table
Oct 7–19	Train fallback B4; first QLoRA run of HUM (S2, 3B then 7B); first LOGO Δ numbers	Oct 20: Checkpoint I (G1)
Oct 20–Nov 2	Add gating fusion and all loss terms; GRL head (S3); pseudo-domain clustering; optional S1	Fused model on LOGO
Nov 3–9	The first five ablations; calibration prototype with λ tuning; demo prototype	Nov 10: Checkpoint II + 15-min talk
Nov 10–18	Remaining ablations; freeze method; dry-run submission on validation	Frozen configuration
Nov 19–Dec 2	Nov. 19 evaluation release: frozen inference, optional calibration, sanity checks; draft system description	Final predictions
Dec 1–8	Submit to XACLE (Dec. 3); final presentation (Dec. 1/3/8); record demo video	Dec 3: XACLE submission
Dec 9–11	Final report, repo tag, video upload, peer evaluations	Dec 11 (8 PM): Final report

APPENDIX C ACKNOWLEDGMENT

This proposal was prepared with the assistance of AI tools which were used to help generate, edit, and organize content and to check references. AI coding assistants will also be used to help assist writing code during the project. All technical choices, claims, citations, and code are reviewed and verified by the team, and the team takes full responsibility for the work.